

Appendix E: Architecture & Matrices

Throughout the book, we discuss complex technical architectures, processing pipelines, and comparison matrices. For readability, the full-page Excalidraw diagrams for these concepts are collected here and hosted in the [building-with-llms-companion](#) repository.

You can view the high-resolution `.png` exports or download the editable `.excalidraw` source files to modify them for your own presentations and design docs.

Diagram Inventory

Chapter 1: The Transformer Architecture

File: [ch01-transformer-full.png](#) A complete visual breakdown of the Transformer architecture, including input embeddings, positional encoding, multi-head attention (Q/K/V), feed-forward layers, and output probabilities.

Chapter 4: Model Capability Matrix

File: [ch04-capability-matrix.png](#) A full capability matrix mapping 10 common generative AI tasks against their reliability rating, the best-in-class model for that task, and the associated cost tier.

Chapter 6: Prompting Technique Matrix

File: [ch06-technique-matrix.png](#) A detailed comparison of 6 advanced prompting techniques (e.g., Zero-Shot, Few-Shot, Chain-of-Thought, ReAct), detailing when to use them, pros, cons, and an example task.

Chapter 8: Evaluation Pipeline

File: [ch08-eval-pipeline.png](#) A visualization of the complete evaluation pipeline: taking a Golden Dataset, passing it through a Prompt Runner, scoring it with an LLM-as-Judge, and finally generating a metrics report.

Chapter 10: Chunking and Indexing Pipeline

File: [ch10-chunking-indexing.png](#) A structural flow of the document ingestion process, tracing the path from raw documents (PDF/Text) to Document Loaders, Text Splitters, Embedding Models, and ultimately the Vector Database.

Chapter 11: RAG Architecture (Full)

File: [ch11-rag-full.png](#) The complete Retrieval-Augmented Generation (RAG) architecture, bridging the offline ingestion pipeline with the real-time query pipeline and the evaluation feedback loops.

Chapter 13: Cost Optimization Decision Tree

File: [ch13-cost-decision-tree.png](#) A practical decision tree for cost optimization, branching through prompt shortening, caching implementation, and model tier switching, with estimated savings percentages at each leaf.

Chapter 14: Production Evaluation Harness

File: [ch14-eval-harness.png](#) A robust production evaluation harness architecture, flowing from Golden Datasets to Multi-Scorers, Drift Monitors, and alerting systems (PagerDuty/Slack).